

SECURE NETWORK WITH FIREWALLS NAT, PORT FILTERING, PACKET FILTERING, AND OTHER METHODS

Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

ANS:

Open `wf.msc` via `Win + R`.

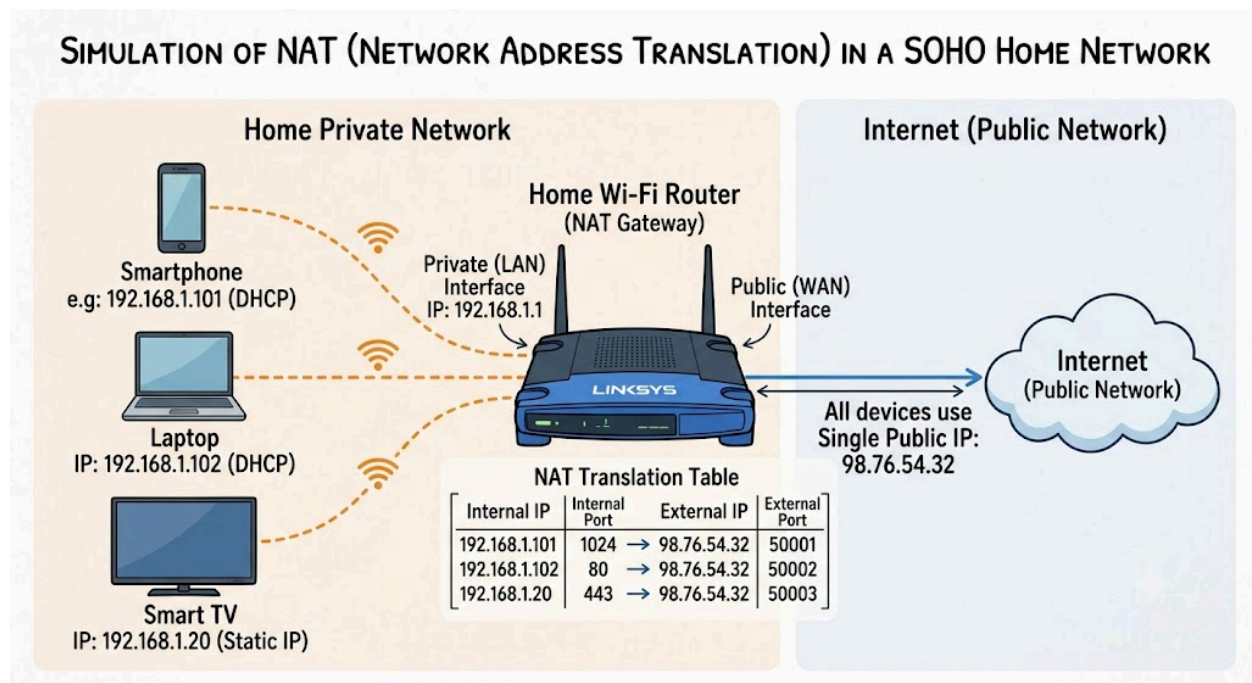
Click **Inbound Rules > New Rule...**

Select **Port > TCP > Specific local ports: 8080**.

Choose **Block the connection** > check all profiles > name it **Block Port 8080**.

Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.
Hint: Use any drawing tool or even hand-draw and click a photo.

ANS:



Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

ANS:

Port Filtering Steps (Port 25 / SMTP)

1. Log into your Wi-Fi router's admin panel (192.168.1.1 or 192.168.0.1).
2. Go to **Security / Advanced > Port Filtering** (or **Outbound Rules / Access Control**).
3. Create a new rule:
 - o **Protocol:** TCP
 - o **Port Range:** 25 (Outbound/Destination Port)
 - o **Source IP:** All local IP addresses (192.168.1.1 – 192.168.1.254)
 - o **Action:** Deny / Block
4. Click **Apply** or **Save** to activate the rule.

Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.

ANS:

Share the trainer's rule set table to get exact results! Based on standard firewall defaults:

1. **HTTP (192.168.1.5 → 172.217.166.78:80): ALLOWED** (standard web browsing)
2. **SSH (10.0.0.10 → 34.201.23.1:22): BLOCKED** (port 22 restricted by default)
3. **DNS (192.168.1.7 → 8.8.8.8:53): ALLOWED** (domain name lookup)

Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

ANS:

DDoS Protection & Rate Limiting (WAF): Drops high-volume packet floods targeting APIs or login endpoints before servers crash.

Port Whitelisting (NACLs): Blocks all public inbound traffic except port 443 (HTTPS), keeping database and SSH ports hidden.

Payload Inspection (DPI): Scans incoming packet content to filter out SQL Injection (SQLi) and Cross-Site Scripting (XSS) attacks.